

# Ing. Michal Štěpán

AI & Data Platform Architect · Senior GenAI Engineer · Azure & GCP

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Portfolio [mxstex.github.io/MyWebPage](https://mxstex.github.io/MyWebPage)

## PROFILE

AI and data platform engineer with about 20 years across physics research, energy, software, data and AI. At ABB I design and run production GenAI: a RAG pipeline on Azure AI Search, the internal platform GenAI applications run on, and Document Intelligence pipelines. Before that, three years as a freelance data and backend engineer for DTSE, Landis+Gyr and Vendavo, and eight years at a district-heating utility, the last ones leading a team of more than fifteen people. Physics background: nuclear engineering at ČVUT FJFI and tokamak research. In my own time I run Tojin Games, where I build and ship simulation games, LLM agents and cloud services end to end. Moving towards AI and data platform architecture and technical leadership.

## CORE SKILLS

### Architecture & GenAI

RAG (hybrid search, grounding, citations) · multi-step LLM workflows · function calling / tools · Model Context Protocol (MCP) · agent systems (LangChain, LangGraph) · model evaluation with golden datasets · local LLMs (Ollama)

### Cloud & infrastructure

Azure (Container Apps, Functions, Storage, Key Vault, Container Registry, Log Analytics) · GCP (Cloud Run, GCS, BigQuery, Secret Manager) · Docker · Terraform · OAuth 2.0 / JWT · Azure Pipelines · GitLab CI · GitHub Actions

### Data & AI platforms

Azure AI Search · Azure OpenAI · Azure Document Intelligence · pgvector · ChromaDB · Dagster · Airflow · PostgreSQL · DuckDB · BigQuery · RabbitMQ · data quality & observability

### Programming

Python (FastAPI, Pandas, NumPy/SciPy, Pytest, Playwright, Streamlit, NiceGUI) · SQL · JavaScript (Canvas, WebGL2) · Kotlin + Jetpack Compose · Grafana · Power BI

## PROFESSIONAL EXPERIENCE

### Senior GenAI Engineer · ABB s.r.o.

Sep 2025 – present

- Designed the architecture of a full RAG pipeline: Azure AI Search, vector embeddings, hybrid retrieval and grounding.
- Integrate GenAI into business processes; build golden datasets and model-evaluation workflows.
- Develop the internal GenAI application platform: FastAPI, Streamlit, Docker, Azure Container Apps.
- Run the platform: Key Vault, Container Registry, Log Analytics, containerized deployments.
- Implement Document Intelligence and multimodal processing pipelines.

Stack: Python · FastAPI · Streamlit · Azure OpenAI · Azure AI Search · Document Intelligence · Docker · Azure Container Apps

### Freelance Software / Data Engineer

Sep 2022 – Sep 2025

DTSE, Landis+Gyr, Vendavo and other clients in energy, telecom and manufacturing · remote, international teams

- Designed and built end-to-end data workflows in Python, orchestrated with Dagster and Airflow, stored in PostgreSQL, DuckDB and BigQuery.
- Built backend components for high-volume ingestion and message-based processing with RabbitMQ, and REST services.
- Migrated services from Kubernetes to cloud-native GCP (Cloud Run, GCS, BigQuery).
- Maintained CI/CD in GitLab CI and GitHub Actions with unit, integration and end-to-end tests in Pytest.
- Delivered high-performance processing pipelines for recommender systems (multiprocessing, optimisation); L2 support for production systems.

Stack: Python · Dagster · Airflow · PostgreSQL · DuckDB · BigQuery · RabbitMQ · GCP · Kubernetes · GitLab CI · GitHub Actions

### Senior Software Developer · Siemens s.r.o.

Jan 2021 – Aug 2022

- Developed an Energy Management System consolidating and analysing data from many sources.
- Implemented ARIMA and SARIMA models for time-series forecasting.
- Mentored junior developers.

Stack: Python · Flask · FastAPI · Docker · PostgreSQL · Grafana · Power BI · RabbitMQ · AWS S3 · AWS Lambda

### Software Project Engineer · ABB s.r.o.

Oct 2018 – Jan 2021

- Developed a water and energy consumption evaluation system; designed and implemented predictive-analytics features.

Stack: Python · Docker · PostgreSQL · Azure Functions · Azure Blob Storage · Grafana · Power BI

- As technical deputy, led an operations team of more than fifteen people in water and heat distribution.
- Improved resource efficiency through optimised calculations and monitoring; streamlined data collection with PDA barcode readers (MS SQL, VBA).

TOJIN GAMES · OWN STUDIO, IN MY OWN TIME

Independent science, space and simulation games · YouTube youtube.com/@tojin111 · live builds on the portfolio site

- Built nine science, space and strategy games solo; five are live in the browser on Google Cloud Run. Hand-written simulation engines for Newtonian gravity, Kepler orbital mechanics with WebGL2 rendering, plasma confinement and the Schrödinger equation; golden-trace tests keep the Python, JavaScript and Kotlin engines in agreement.
- **Jednota**, a multi-agent LLM strategy game (~73,000 lines of Python, 1,300 tests): strategist, envoy and admiral agents on local LLMs through Ollama; deterministic rules validate every model decision (accepted 93/93 strategic, 80/88 diplomatic, 79/86 tactical in a measured campaign); memory across games through pgvector RAG.
- **Platform**: an OAuth 2.0 to RS256 JWT entitlement service (FastAPI, Cloud Run, Secret Manager, Terraform, 54 tests) gives five games one shared login and access tiers; every service scales to zero under a one-instance cost cap.
- **Video pipeline**: code-driven 4K capture of real game sessions (Playwright, Chrome DevTools Protocol, ffmpeg/NVENC), captions from on-screen events, local TTS, loudness-checked masters and automated validation, plus local ComfyUI models (FLUX, Wan 2.2): more than forty videos published in September 2026.
- Built with AI coding agents (Claude Code) under automated test gates: level verifiers, bots that play the games through, cross-language golden traces.

EDUCATION, CERTIFICATES, LANGUAGES

Ing. (MSc), Electrical Engineering and Computer Science · 2012

VŠB – Technical University of Ostrava. Thesis: VBA Utilization for Power Engineering Data Processing and Presentation

Bc., Nuclear Engineering · 2009

Czech Technical University in Prague, Faculty of Nuclear Sciences and Physical Engineering. Thesis: Physical Analysis of Fast Reactor Cores in Up-to-Date Tendencies

Research project · 2006

Institute of Plasma Physics, Czech Academy of Sciences: electrostatic turbulence in the edge plasma of the CASTOR tokamak

Microsoft AZ-305

Designing Microsoft Azure Infrastructure Solutions · training

Microsoft AZ-400

Designing and Implementing Microsoft DevOps Solutions · training

Cambridge English: First (FCE)

Grade B · CEFR B2 · 2019

Czech

native

English

B2–C1 (Cambridge FCE)